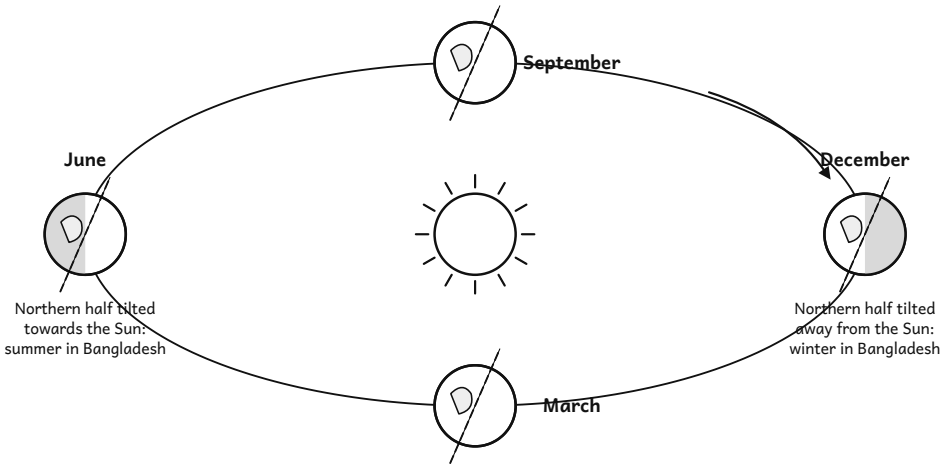
EARTH AND SPACE

Earth, Sun and Moon: Days, Years and Seasons

Day and night, the Sun's path, years, seasons around the world, the Moon, planets and stars



In this pack

📖 **Learn it:** Earth, Sun and Moon: days, years and seasons ★☆☆
Fact check: Days, years and planets ★☆☆
Think it through: Explaining what we see ★★☆
Investigate: How does day length change through the year? ★★☆
Real-world case study: Six seasons, one planet ★★★
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This pack belongs to: _____ **Class:** _____

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Earth, Sun and Moon: days, years and seasons

★☆☆ Grade 5

Name: Class: Date:

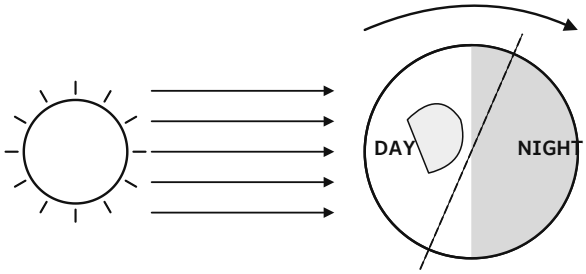
Learning goals

- Explain day and night and the apparent movement of the Sun.
- Explain why different places have different seasons.
- Describe the solar system and how scientists came to understand it.

Every morning the Sun seems to rise in the **east**, climb across the sky and set in the **west**. For thousands of years people believed the Sun travelled around the Earth. Today we know it is the **Earth that moves**.

Day and night: the Earth spins

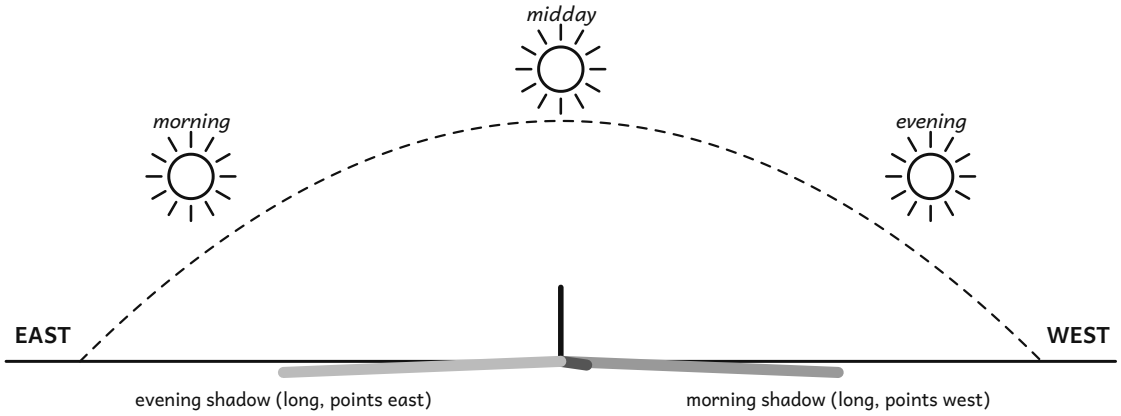
The Earth spins on an imaginary line through the North and South Poles called its **axis**. One complete spin (a **rotation**) takes **24 hours**. The half of the Earth facing the Sun has **day**; the half facing away has **night**. Because the Earth spins from west to east, the Sun *appears* to move from east to west.



The Earth spins on its axis once every 24 hours.

The Sun's path and shadows

In the morning the Sun is low in the east and shadows are long, pointing west. At midday the Sun is highest and shadows are shortest. In the evening shadows are long again, pointing east.

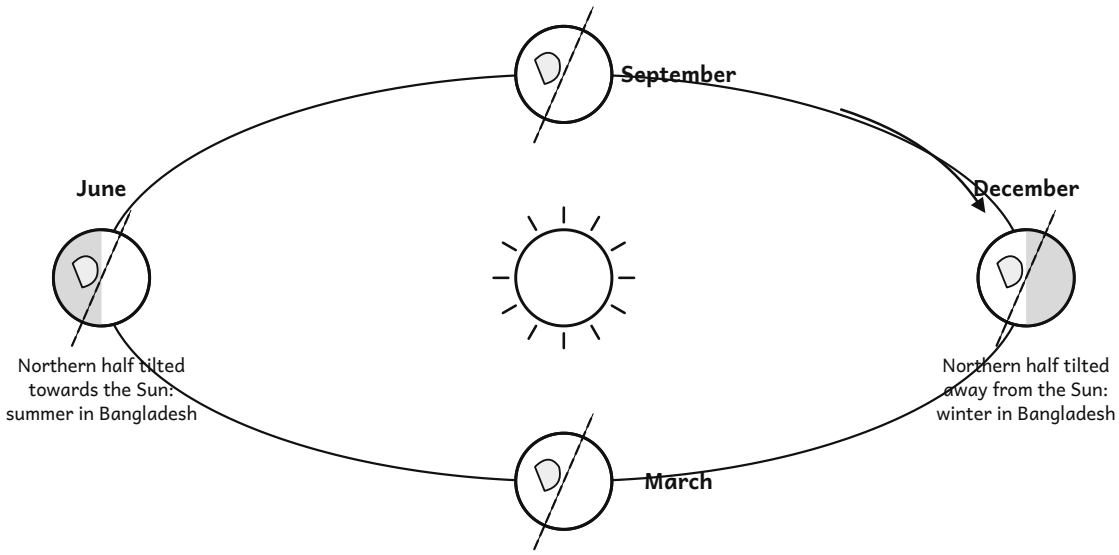


A year: the Earth orbits the Sun

The Earth travels around the Sun in a path called an **orbit**. One orbit takes about **365¼ days**, which is one **year**. The extra quarter day adds up to one extra day every four years: a **leap year** with 29 February.

Why are there seasons?

The Earth's axis is **tilted** by about 23.5°. As the Earth orbits, the northern half is tilted **towards** the Sun for part of the year: sunlight hits it more directly, days are longer, and it is **summer**. Six months later it is tilted ***away***: sunlight is spread out, days are shorter, and it is **winter**. When it is summer in Bangladesh, it is winter in Australia!



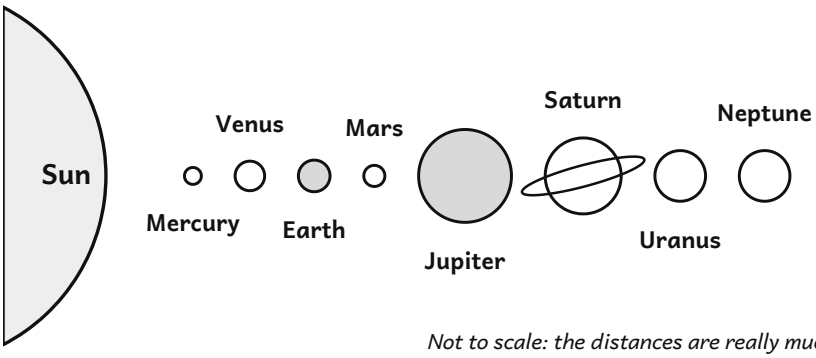
Places near the **equator**, such as Singapore, have about 12 hours of daylight all year and no real summer or winter. Places near the **poles** have the biggest changes: in summer the Sun never sets (the "midnight sun"), and in winter it never rises. Bangladesh is near the **Tropic of Cancer**, which actually passes through the country.

The Moon

The Moon orbits the Earth about once a month. It does not make its own light: we see it because it **reflects** sunlight. As it orbits, we see different amounts of its lit half, so its shape seems to change. These are its **phases**, from new moon to full moon and back in about 29½ days. The Islamic calendar follows the Moon.

The solar system

The Sun is a **star**. Eight **planets** orbit it: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. A sentence can help you remember the order: "**M**y **V**ery **E**asy **M**ethod **J**ust **S**peeds **U**p **N**aming".



Not to scale: the distances are really much, much bigger.

Stars and how scientists learned

Stars are huge, hot balls of gas, like the Sun but much further away. Groups of stars that make patterns are **constellations**. In Bangla, Orion is *Kalpurush* and the Great Bear's seven bright stars are *Saptarshi Mandal*. Around 1500 years ago the Indian astronomer **Aryabhata** suggested the Earth spins. In 1543 **Copernicus** argued that the Earth goes around the Sun, and in 1610 **Galileo** used a telescope to see moons orbiting Jupiter, which was strong evidence that not everything orbits the Earth.

Key words

axis	imaginary line the Earth spins around
rotation	one complete spin (24 hours)
orbit	the path of one object around another
tilt	the lean of the Earth's axis (about 23.5°)
equator	imaginary line around the middle of the Earth
phases of the Moon	the changing shapes of the Moon we see
planet	a large body that orbits a star
constellation	a pattern of stars

FACT CHECK · B4 · EARTH AND SPACE

Days, years and planets

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Recall facts about the Earth, Sun and Moon.
- Order the planets.

1 **Fill in the gaps.** [6]

WORD BANK

24 **12** **reflects** **east** **365¼** **axis** **west** **north**

The Earth spins on its _____ once every _____ hours, causing day and night. It orbits the Sun once every _____ days. The Moon _____ light from the Sun. The Sun rises in the _____ and sets in the _____.

2 **Number the planets in order from the Sun (1 = closest).** [4]

- ☐ Mars
- ☐ Uranus
- ☐ Mercury
- ☐ Venus
- ☐ Saturn
- ☐ Earth
- ☐ Jupiter
- ☐ Neptune

3 True or false? [5]

Statement	True	False	If false, correct it
1. The Sun moves around the Earth every day.			
2. The Moon makes its own light.			
3. When it is summer in Bangladesh it is winter in Australia.			
4. Shadows are shortest at midday.			
5. The Sun is a star.			

4 Match each word to its meaning. [4]

Draw a line to match each pair, or write the letter.

rotation ●	● **A** one spin of the Earth on its axis
orbit ●	● **B** a year with 366 days
leap year ●	● **C** the path of the Earth around the Sun
constellation ●	● **D** a pattern of stars

Explaining what we see

★★☆ Grade 5

Name: Class: Date:

Learning goals

- Explain day, night and seasons with diagrams.
- Use models.

1 **Shade the night side of the Earth and label day and night.** [2]

Sun on the left, Earth on the right

2 **Explain why it is daytime in Dhaka when it is night-time in New York.** [2]

3 **Explain why Bangladesh is hotter in June than in December. Use the words *tilted*, *direct* and *longer*.** [3]

4 **A model: you shine a torch (Sun) at a ball (Earth) with a stick through it (axis) and turn the ball.** [3]

a) What does turning the ball show?

b) Give **one** way this model is not like the real Earth and Sun.

5 **Why does the Moon seem to change shape?** [2]

How does day length change through the year?

★★☆ Grade 5

Name: Class: Date:

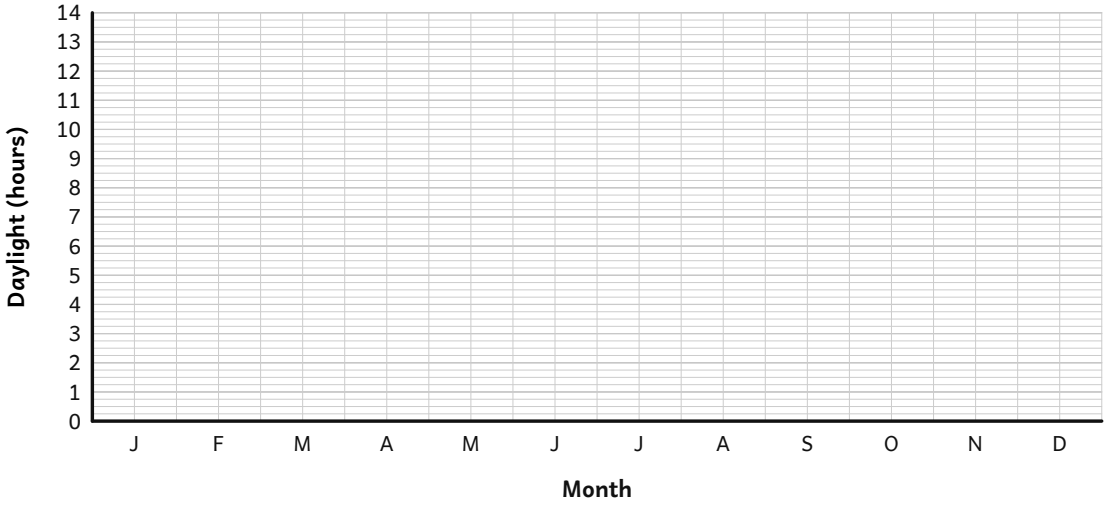
Learning goals

- Plot a line graph.
- Identify a pattern.
- Explain it using the Earth's tilt.

These are approximate hours of daylight in **Dhaka** in the middle of each month.

Month	J	F	M	A	M	J	J	A	S	O	N	D
Daylight (hours)	10.9	11.4	12.0	12.7	13.3	13.6	13.4	12.9	12.2	11.6	11.0	10.7

1 **Plot a line graph of daylight hours.** [4]



2 **Answer using your graph.** [4]

a) Which month has the longest days? the shortest?

.....

b) Describe the pattern.

.....

.....

c) How much longer is a day in June than in December?

.....

3 **Predict the daylight hours in Singapore (on the equator) in June and in December. Explain.** [2]

.....

.....

4 **Shadow stick. Plan how you would measure the shadow of a stick at different times of day. What would you keep the same?** [2]

REAL-WORLD CASE STUDY · B4 · EARTH AND SPACE

Six seasons, one planet

★★★
Grade 5

Name: **Class:** **Date:**

Learning goals

- Compare seasons in different places.
- Link seasons to culture and farming.
- Explain using the Earth's tilt and position.

Science · Geography · Culture · Maths · Agriculture

Bengali tradition divides the year into **six seasons** (*ritu*): *Grishmo* (summer), *Borsha* (monsoon), *Shorot* (early autumn), *Hemonto* (late autumn), *Sheet* (winter) and *Boshonto* (spring). Farming follows the seasons: *Aman* rice grows in the monsoon and is harvested in Hemonto, celebrated with the **Nabanna** festival of new rice, while *Boro* rice grows in the dry winter with water from tube-wells.

Other places have very different seasons. In **Singapore**, near the equator, it is hot all year and every day has about 12 hours of daylight. In **Tromsø**, Norway, far to the north, the Sun does not set for about two months in summer and does not rise for about two months in winter. In **Sydney**, Australia, south of the equator, December is the middle of summer.

Day-length facts from standard astronomical tables (approximate).

1 **Complete the table.** [6]

Place	Where it is	What its seasons are like
Dhaka		
Singapore		
Tromsø		
Sydney		

2 **Explain why Sydney has summer in December, when Dhaka has winter.** [2]

3 **Choose one Bangladeshi festival or food linked to a season. Explain the link.** [2]

4 **Why does Boro rice need water from tube-wells, but Aman rice does not?** [2]

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

R	S	U	N	W	P	Y	G	L	J
T	T	A	P	R	M	R	K	G	I
E	M	M	I	N	W	S	T	E	T
Q	Y	B	O	R	B	I	T	R	B
U	F	Y	C	O	H	I	B	W	J
A	R	G	F	W	N	O	F	C	H
T	I	L	T	T	M	S	T	A	R
O	J	L	P	L	A	N	E	T	Y
R	G	H	S	E	A	S	O	N	H
K	L	A	X	I	S	R	D	J	M

axis **equator** **moon** **orbit** **planet** **season** **star** **sun** **tilt**

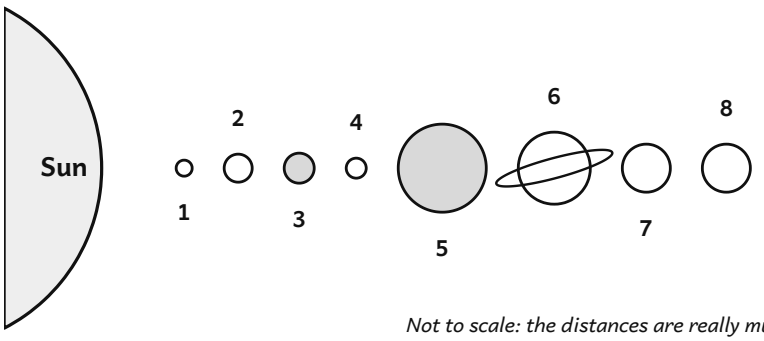
2 **What causes day and night?** [1]

- ◯ the Earth orbiting the Sun
- ◯ the Earth spinning on its axis
- ◯ the Moon blocking the Sun
- ◯ clouds

3 **What causes the seasons?** [1]

- ◯ the Earth's distance from the Moon
- ◯ the tilt of the Earth's axis as it orbits the Sun
- ◯ the Sun getting hotter
- ◯ the wind

4 **Label the planets with numbers 1 to 8 from the Sun, then write the name of the largest planet.** [2]



Not to scale: the distances are really much, much bigger.

5 **Challenge. Galileo saw moons going around Jupiter. Why was this important evidence?** [2]

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain day and night	○	○	○
explain why the Sun appears to move across the sky	○	○	○
explain why we have seasons	○	○	○
explain why different places have different seasons	○	○	○
name the planets in order	○	○	○
plot and read a line graph of day length	○	○	○